

Carlos Moreno Rojas

Santiago, Chile | camoro2002@uc.cl

RESEARCH PROFILE

Biomedical and electrical engineer and M.Sc. candidate working at the interface of **biomedical instrumentation, organ perfusion, embedded systems, signal processing, and NIR fluorescence imaging**. Research focuses on translating physiological questions into measurable engineering variables, building reproducible experimental platforms, and validating complete hardware-software systems for biomedical research.

RESEARCH INTERESTS

Biomedical instrumentation; organ perfusion and preservation; liver viability assessment; ICG/NIR fluorescence; biosensors; physiological monitoring; embedded systems and data acquisition; biomedical signal and image processing; medical-device prototyping; interdisciplinary experimental engineering.

EDUCATION

Pontificia Universidad Católica de Chile 2025–2027 (*expected*)
M.Sc. Candidate in Biomedical Engineering – Biosensors / Biomedical Instrumentation Santiago, Chile

- Supervisor: **Dr. Rolando Rebolledo**.
- Thesis: *Development of an experimental platform for ex vivo liver viability assessment using ICG-NIR fluorescence.*

Pontificia Universidad Católica de Chile 2025–2026
Professional Degree: Ingeniero Civil Eléctrico (Electrical Engineering) Santiago, Chile

Pontificia Universidad Católica de Chile 2021–2024
Bachelor's Degree in Engineering Sciences; Major in Biomedical Engineering, Minor in Biomedical Engineering Santiago, Chile

RESEARCH EXPERIENCE

Master's Thesis Research – Ex Vivo Liver Viability and ICG-NIR 2025–Present
Pontificia Universidad Católica de Chile – Biomedical Engineering Santiago, Chile

- Redesigned an initially unstable liver-perfusion setup into a more reproducible experimental platform for an **ex vivo rat-liver model**, integrating pressure, flow, temperature, oxygenation, pulse damping, and bubble management.
- Developed a modular **hydrodynamic monitor and data-logging architecture** for pressure, flow, temperature, and vascular resistance using clinical sensors, embedded electronics, and real-time acquisition software.
- Calibrated and validated measurements against certified references, including a **Fluke ProSim 8**, using correlation, deviation, and RMSE; implemented sampling and filtering strategies to distinguish real dynamics from acquisition noise.
- Designed the optical measurement strategy for **ICG fluorescence in the near-infrared**, including NIR illumination, emission filtering, camera selection, RAW acquisition, dark-chamber control, and concentration-based calibration.
- Investigating quantitative ICG kinetics (e.g., peak, time-to-peak, elimination slope, k , PDR, R15, AUC, washout) and their relationship to independent physiological markers toward a predictive model of liver viability.

Organ Preservation and Supercooling – Borealis UC 2024
Pontificia Universidad Católica de Chile – Dr. Rolando Rebolledo Santiago, Chile

- Contributed to a project investigating **supercooling-based organ preservation** to reduce hypoxic damage during transport and extend preservation time for transplantation.
- Continued this research direction through electromechanical redesign and miniaturization of an organ-preservation machine, integrating mechanics, electronics, control, fluids, and experimental testing.

Medical Image Segmentation with U-Net 2025
Pontificia Universidad Católica de Chile – Dr. Marcelo Andia Santiago, Chile

- Developed a **U-Net-based neural network** for lung segmentation in CT and MRI, applying deep learning to biomedical image analysis.

SELECTED ENGINEERING AND BIOMEDICAL PROJECTS

Candel Medical Company – Hardware/Firmware Engineering Intern

Jan.–Mar.

Transcranial Electrical Stimulation (tES)

Chile

- Improved a medical-device prototype toward product-level integration through **circuit redesign, PCB debugging, firmware development, and experimental validation**.
- Worked with **ESP32 and STM32**, BLE communication, signal generation and synchronization, digital control, power reduction, and hardware-software integration for tDCS/tACS/tRNS stimulation modes.

Ciervo UC – Electrical Area Leader, CYBATHLON 2024

2024

Robotic Transfemoral Prosthesis; ETH Zürich CYBATHLON

Santiago, Chile / Zürich, Switzerland

- Led electrical/electronic development for a **robotic transfemoral prosthesis**, including prototype design and manufacturing, system integration, and active-system development.
- Integrated electronics with control and mechanical subsystems under real operational constraints; the system was taken to the international CYBATHLON 2024 competition in Zürich.

FPGA-Based Upgrade for Universal Testing Machine

2025–2026

Meldic Ltda.

Santiago, Chile

- Developed an embedded processing architecture using **Zybo Z10, Vivado, and VHDL** for force, displacement, deformation, and speed-control signals in mechanical testing equipment.

3D Design for GE Vscan Air Accessory

2023

Institute for Biological and Medical Engineering (IIBM UC) – Dr. Martín Miranda

Santiago, Chile

- Designed a 3D cover/enclosure for the **GE Vscan Air** handheld ultrasound transducer using CAD-based prototyping.

TEACHING AND LEADERSHIP

Laboratory Teaching Assistant – FIS154

Apr.–Jun. 2024

Faculty of Physics, Pontificia Universidad Católica de Chile

Santiago, Chile

- Managed and supervised laboratory activities, supported students, and communicated experimental procedures and technical concepts.

President – Biomedical Engineering Student Chapter UC

2024–2025

Pontificia Universidad Católica de Chile

Santiago, Chile

Webmaster – IEEE EMBS PUC Chile

2024–Present

IEEE Engineering in Medicine and Biology Society

Santiago, Chile

TECHNICAL SKILLS

Biomedical / Experimental: organ perfusion, physiological monitoring, ICG fluorescence, NIR imaging, biosensors, medical-device prototyping, validation, calibration, experimental design.

Electronics / Embedded: analog and digital electronics, ESP32, STM32, Arduino, ADCs, HX711, PT100/PT1000, clinical pressure sensors, ultrasonic flow sensing, BLE, PCB debugging, instrumentation.

FPGA / Digital Design: Vivado, VHDL, Zybo Z10, embedded signal-processing architectures.

Software / Data: Python, C/C++, acquisition software, signal processing, image processing, neural networks, U-Net, data analysis, Markdown.

Optics / Imaging: NIR cameras, optical filters, irradiance, quantum efficiency, full-well capacity, SNR, RAW image acquisition, fluorescence calibration.

Design / Fabrication: Fusion 360, CAD, 3D printing, CNC/machining concepts, prototyping, electromechanical integration.

LANGUAGES

Spanish: Native

English: B2 / ALTE 3 (TOEIC)